

Task-05 Malware Types and Behaviour Analysis

Chapter 1: Introduction to Malware and Malware Analysis

1.1 What is a Malware?

Malware is short for *malicious software*. It refers to any program or code that is intentionally designed to harm a computer system, network, or user data. Unlike normal software, malware is created with the purpose of causing damage, stealing information, or gaining unauthorized access to systems.

Common goals of malware include:

- Stealing sensitive information such as passwords or personal data
- Disrupting system operations
- Encrypting files for ransom
- Spreading itself to other systems without user consent

Malware often operates silently in the background, making it difficult for users to detect until significant damage has already occurred.

1.2 Why Malware Analysis is Important

Malware analysis is the process of studying malicious software to understand **what it does**, **how it works**, and **how it can be detected or prevented**. In cybersecurity, malware analysis plays a critical role because it helps security professionals respond effectively to attacks.

By analyzing malware, we can:

- Identify the type of malware involved
- Understand its behavior and impact on a system
- Detect how it spreads
- Develop security measures to stop similar attacks in the future

Without malware analysis, organizations would only be reacting to attacks instead of proactively defending against them.

1.3 Role of Malware Analysis in Cybersecurity

In real-world cybersecurity, malware analysis helps:

- Antivirus vendors improve detection signatures
- Security teams identify Indicators of Compromise (IOCs)
- Organizations respond faster to incidents
- Researchers understand emerging attack techniques

Tools like **VirusTotal** allow analysts to examine malware samples safely by scanning them across multiple security engines and analyzing their behavior without executing them on a live system.

1.4 Scope of This Task

In this task, malware analysis is performed at a **basic level** using **VirusTotal**. Instead of executing malware files directly, known malware **hash values** are analyzed to study detection results and behavioral indicators.

The ransomware **WannaCry** is used as the primary case study for this analysis. By examining WannaCry, this task focuses on:

- Understanding malware classification
- Observing detection reports from multiple vendors
- Analyzing behavioral characteristics
- Learning prevention and mitigation techniques

This approach ensures safe and effective learning while providing real-world exposure to malware analysis concepts.

Chapter 2: Common Malware Types

Malware can be classified into different types based on **how it spreads, what it does, and the damage it causes**. Below are some of the most observed malware categories in cybersecurity.

1. Virus

A **virus** is malware that attaches itself to a legitimate file or program and spreads when that file is executed.

Key characteristics:

- Requires user action to spread (e.g., opening a file)
- Can corrupt or modify files
- Often spreads via removable media or email attachments

Example impact: File corruption, system instability

2. Worm

A **worm** is a standalone malware that can spread automatically across networks without user interaction.

Key characteristics:

- Exploits network or system vulnerabilities
- Spreads rapidly across connected systems
- Consumes bandwidth and system resources

Example impact: Network congestion, large-scale infections

Example: WannaCry (used worm-like behavior)

3. Trojan

A **Trojan horse** disguises itself as legitimate software to trick users into executing it.

Key characteristics:

- Does not self-replicate
- Often creates backdoors

- Used to deliver other malware

Example impact: Unauthorized access, data theft

4. Ransomware

Ransomware encrypts a victim's files and demands payment to restore access.

Key characteristics:

- Uses strong encryption
- Causes immediate operational disruption
- Often spreads via phishing or exploits

Example impact: Data loss, financial damage

Example: WannaCry ransomware

5. Spyware

Spyware secretly monitors user activity and collects sensitive information.

Key characteristics:

- Runs silently in the background
- Tracks keystrokes, browsing habits, or credentials
- Often bundled with free software

Example impact: Privacy invasion, credential theft

6. Adware

Adware displays unwanted advertisements and may track user behavior for marketing purposes.

Key characteristics:

- Redirects browsers
- Injects ads into webpages
- Sometimes bundled with legitimate applications

Example impact: Poor system performance, privacy risks

7. Rootkit

A **rootkit** is designed to hide malware and maintain persistent privileged access to a system.

Key characteristics:

- Operates at kernel or system level
- Extremely difficult to detect
- Often used to protect other malware

Example impact: Complete system compromise

8. Bot / Botnet Malware

A **bot** turns an infected system into a remotely controlled device, often part of a **botnet**.

Key characteristics:

- Controlled by a command-and-control (C2) server
- Used for DDoS attacks, spam, or credential stuffing
- Infected systems often remain unnoticed

Example impact: Large-scale coordinated attacks

Why Understanding Malware Types Matters

Each malware type:

- Uses **different attack methods**
- Shows **different behavioral indicators**
- Requires **different detection and prevention strategies**

This classification helps analysts interpret **VirusTotal results**, behavioral reports, and detection labels more effectively.

Chapter 3: Malware Detection Using VirusTotal

Detecting malware is not only about identifying whether a file is malicious, but also about understanding **how and why it is classified as a threat**. One widely used platform for this purpose is **VirusTotal**.

What is VirusTotal?

VirusTotal is an **online malware analysis platform** that allows users to analyze files, URLs, and hashes using results from multiple antivirus engines and sandbox environments.

Instead of relying on a single security product, VirusTotal aggregates detection results from **dozens of antivirus vendors**, providing a broader and more reliable view of potential threats.

Why Hash-Based Analysis is Used

In malware analysis, **hash values** (such as MD5 or SHA-256) uniquely represent a file's content. Submitting a hash instead of uploading the actual file is safer and commonly used in academic and professional environments.

Hash-based analysis helps to:

- Avoid executing or downloading malicious files
- Quickly identify known malware samples
- Retrieve historical detection and behavior data
- Analyze malware safely and legally

In this task, a known WannaCry ransomware hash was used to perform analysis without handling the actual malware file.

Detection Reports and Antivirus Engines

When a malware hash is searched on VirusTotal, the platform displays a **detection summary**, which includes:

- Number of security vendors flagging the file as malicious
- Malware categories (e.g., ransomware, trojan)
- Malware family names (e.g., WannaCry)

A high detection ratio indicates that the file is widely recognized as malicious across the cybersecurity community.

Importance of Multi-Engine Detection

Different antivirus engines use different detection techniques, such as:

- Signature-based detection
- Heuristic analysis
- Behavioral analysis
- Machine learning models

By combining results from multiple engines, VirusTotal provides a **more accurate and trustworthy assessment** compared to relying on a single antivirus product.

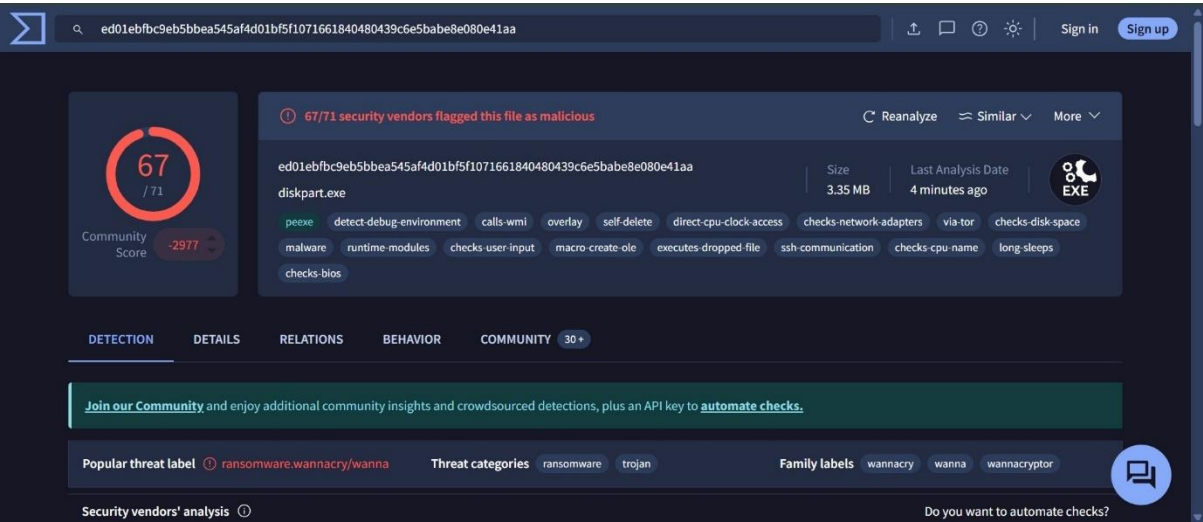
Role of VirusTotal in This Task

VirusTotal was used in this task to:

- Identify the malware type and family
- Analyze detection consistency across vendors
- Observe behavioral indicators linked to ransomware
- Support theoretical understanding with real-world data

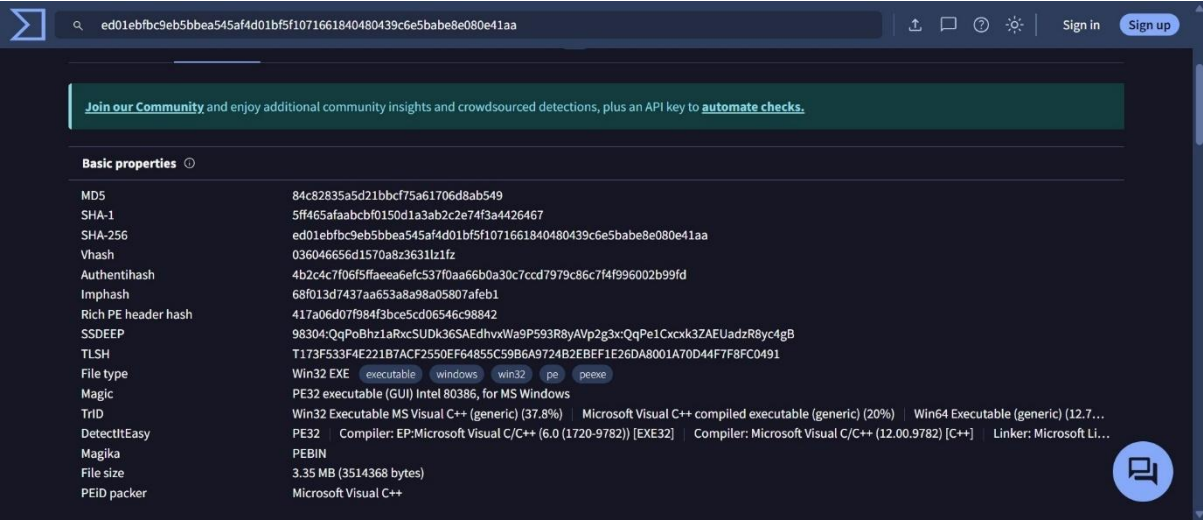
This approach allowed malware analysis to be performed in a **controlled, safe, and educational manner**, aligning with the goals of the task.

Figure 3.1: VirusTotal Detection Report for WannaCry Ransomware



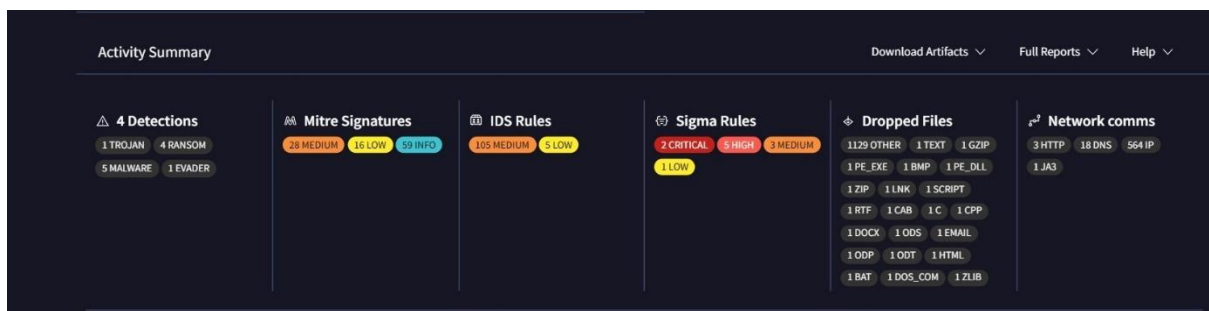
Note: This screenshot shows the detection summary of the WannaCry malware sample, where many security vendors flag the file as malicious. The high detection ratio confirms its classification as ransomware.

Figure 3.2: File Details of WannaCry Malware Sample



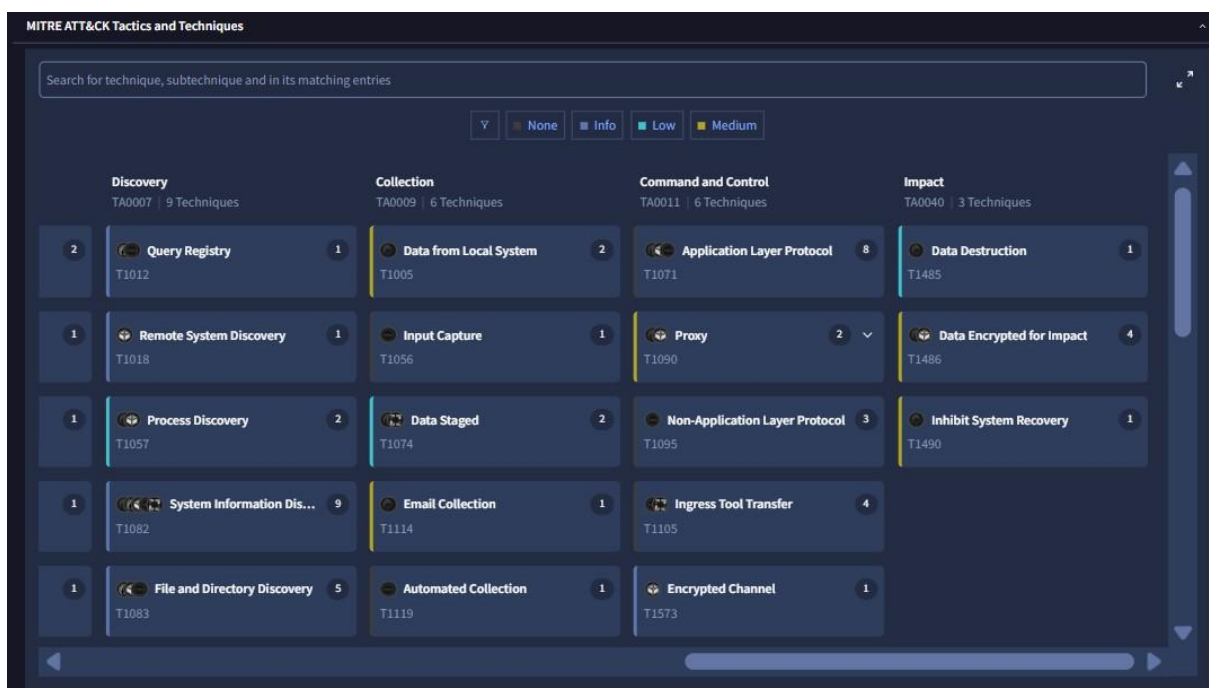
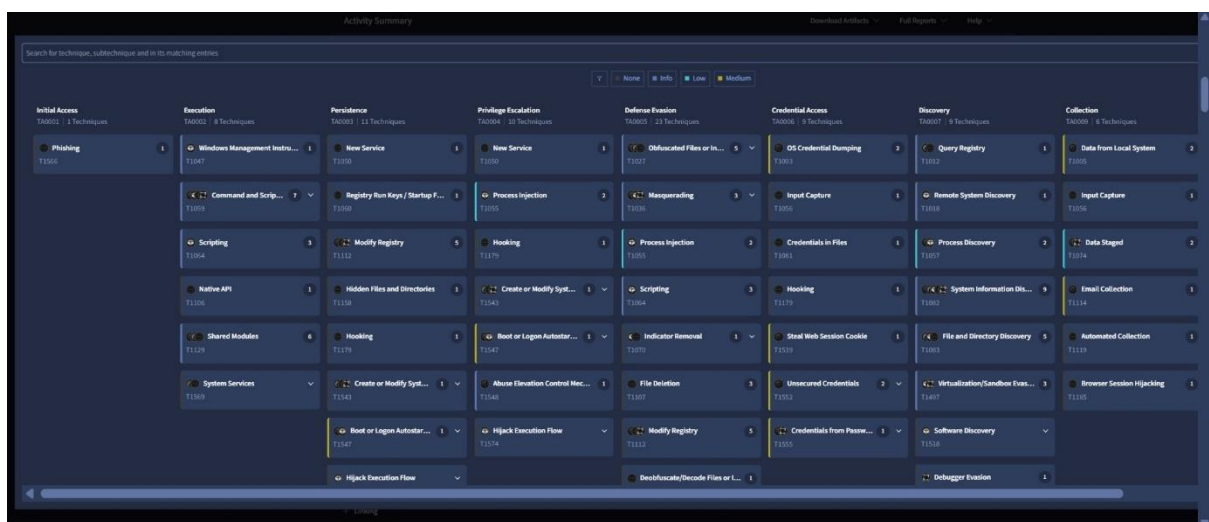
Note: The file details section provides information such as file type, cryptographic hash values, and first-seen date, confirming that the analyzed hash corresponds to a known malware sample

Figure 3.3: Activity Summary of WannaCry Ransomware



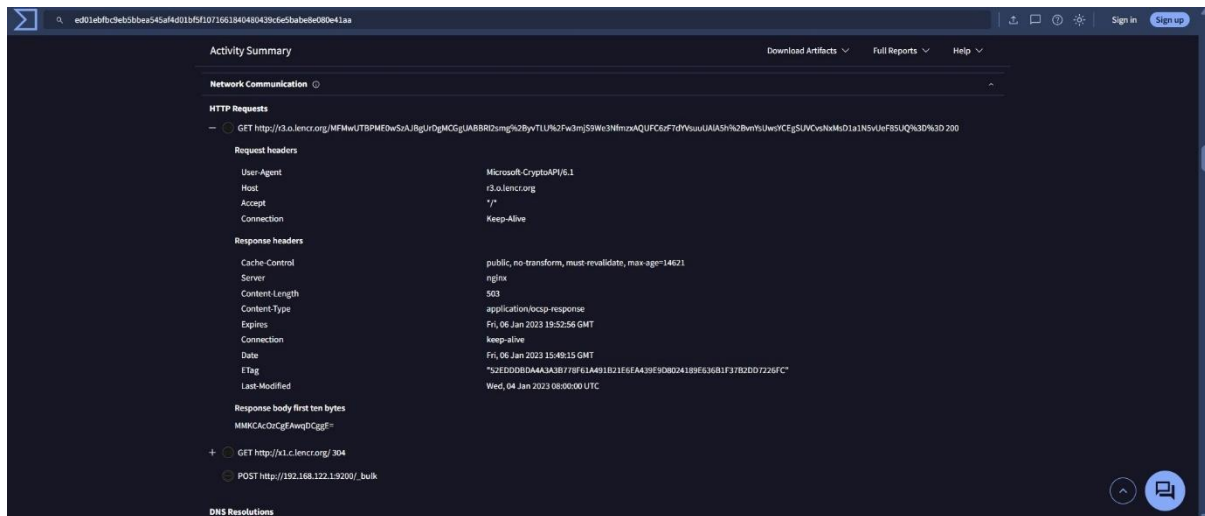
Note: This activity summary highlights the overall malicious behavior observed during analysis, including system-level actions and file-related operations commonly associated with ransomware.

Figure 3.4: MITRE ATT&CK Techniques Associated with WannaCry



Note: This screenshot maps WannaCry's behavior to MITRE ATT&CK techniques, providing insight into the tactics and methods used by the ransomware during execution and propagation.

Figure 3.5: Network Communication Indicators of WannaCry



Note: This screenshot shows network-related indicators such as contacted domains and IP addresses, indicating how the malware attempts to communicate or propagate across systems.

Figure 3.6: Dropped Files, Contacted Domains and Related Artifacts of WannaCry

The figure consists of two screenshots of a malware analysis tool interface, likely VirusTotal, showing artifacts related to the WannaCry ransomware.

Top Screenshot: Contacted Domains (32)

The top screenshot displays the 'Contacted Domains' section for the hash `ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa`. The interface shows a search bar with the hash and a URL `http://192.168.122.1:9200/_bulk`. Below the search bar, there is a table of contacted domains.

Domain	Detections	Created	Registrar
129.214.248.87.in-addr.arpa	0 / 92	-	-
147.251.123.92.in-addr.arpa	0 / 92	-	-
152.251.123.92.in-addr.arpa	0 / 92	-	-
245.50.188.52.in-addr.arpa	0 / 92	-	-
61.234.212.23.in-addr.arpa	0 / 92	-	-
62.102.50.20.in-addr.arpa	0 / 92	-	-
82.250.63.168.in-addr.arpa	0 / 92	-	-
a1666.dscr.akamai.net	0 / 92	1999-03-03	MarkMonitor Inc.
a1672.dscr.akamai.net	0 / 92	1999-03-03	MarkMonitor Inc.
assets.msn.com	0 / 92	1994-11-10	MarkMonitor Inc.

Bottom Screenshot: Dropped Files (100 K)

The bottom screenshot displays the 'Dropped Files' section for the same hash. The interface shows a search bar with the hash and a file name `lhdfgrui.exe`. Below the search bar, there is a table of dropped files.

Scanned	Detections	File type	Name
2023-12-22	12 / 61	Windows shortcut	@WanaDecryptor@.exe.lnk
2025-09-18	0 / 62	DOS EXE	43.WNCRYT (copy)
2026-01-19	0 / 61	JavaScript	166.WNCRYT
2023-09-05	11 / 60	Windows shortcut	@WanaDecryptor@.exe.lnk
2025-10-01	14 / 63	Windows shortcut	@WanaDecryptor@.exe.lnk
2026-01-09	25 / 62	Windows shortcut	@WanaDecryptor@.exe.lnk
2023-05-30	26 / 60	Windows shortcut	@WanaDecryptor@.exe.lnk
2024-01-01	12 / 61	Windows shortcut	@WanaDecryptor@.exe.lnk
2023-09-03	0 / 53	DOS EXE	49.WNCRYT (copy)
2025-09-22	0 / 62	?	116241758512184.bat

Note: This section displays files, domains and artifacts created or dropped by the malware during execution, supporting the behavioral analysis of the ransomware.

Chapter 4: Malware Behavior Indicators

Malware behavior indicators refer to the **observable actions performed by malicious software** when it runs on a system. Instead of focusing only on file signatures or names, behavior analysis looks at *what the malware does*, which is critical for identifying modern and evolving threats.

Common Behavioral Indicators of Malware

Based on general malware analysis practices and the WannaCry sample observed in this task, the following behavior indicators are commonly used to identify malicious activity:

1. File System Activity

Malware often interacts with the file system in suspicious ways, such as:

- Creating or modifying system files
- Dropping additional executable files
- Encrypting user files (common in ransomware)

Observed in WannaCry:

The ransomware is known to encrypt user files, making them inaccessible and demanding payment for decryption.

2. Process Execution and Injection

Malware may:

- Spawn new processes
- Inject code into legitimate system processes
- Run automatically without user knowledge

These actions help malware maintain execution and evade detection.

3. Registry Modifications

Many malware samples modify system registry entries to:

- Achieve persistence

- Execute on system startup
- Alter system configurations

Such changes allow malware to remain active even after system reboots.

4. Network Communication

Malware frequently communicates with external systems to:

- Receive commands
- Exfiltrate data
- Download additional payloads

Observed in WannaCry:

The ransomware attempts network communication, which plays a role in its spreading mechanism and control behavior.

5. Privilege Abuse and System-Level Actions

Some malware attempts to:

- Exploit system vulnerabilities
- Gain elevated privileges
- Disable security features

These actions increase the impact and control malware can have over a compromised system.

Why Behavior Indicators Matter

Behavior-based indicators are especially important because:

- Malware can change its appearance to avoid signature detection
- Behavioral actions are harder to hide
- They allow detection of new or unknown malware variants

By observing runtime behavior, security analysts can **identify threats even when signatures are unavailable.**

Relevance to This Task

The behavior indicators observed in the WannaCry analysis align with known ransomware characteristics, such as file encryption, system modification, and network activity. This confirms the classification provided by VirusTotal and demonstrates how behavior analysis supports malware detection.

Chapter 5: Malware Lifecycle and Spread

Malware does not act randomly. Most malware follows a **structured lifecycle**, moving through different stages from initial infection to execution and persistence. Understanding this lifecycle helps security professionals detect threats early and limit damage.

Malware Lifecycle Stages

1. Initial Infection

This is the stage where malware first enters a system.

Common infection methods include:

- Phishing emails with malicious attachments or links
- Exploiting unpatched system vulnerabilities
- Downloading malicious software disguised as legitimate applications

Example:

WannaCry initially infected systems by exploiting a known Windows vulnerability.

2. Execution

Once inside the system, malware begins execution.

At this stage, malware may:

- Launch malicious processes
- Run scripts or executables
- Start encryption or data manipulation

Execution marks the point where visible damage often begins.

3. Persistence

To survive system restarts, malware attempts to remain active by:

- Modifying registry entries
- Creating scheduled tasks
- Installing itself in startup locations

Persistence allows malware to maintain long-term control over the infected system.

4. Propagation (Spread)

Some malware is designed to spread beyond the initial system.

Propagation methods include:

- Network scanning and exploitation
- Removable media
- Shared drives and services

Observed in WannaCry:

The ransomware used worm-like behavior to spread automatically across vulnerable systems.

5. Payload Delivery

The payload is the **main malicious action** of the malware.

Depending on the type, this may include:

- File encryption (ransomware)
- Data theft (spyware)
- Remote control (botnets)

In ransomware attacks, payload delivery directly impacts system availability.

Why Understanding the Lifecycle Matters

Understanding the malware lifecycle helps:

- Detect malware at early stages
- Apply targeted security controls
- Reduce overall impact of attacks

Breaking the lifecycle at any stage can prevent full system compromise.

Chapter 6: Malware Prevention and Mitigation

Preventing malware infections is more effective than responding after damage has already occurred. Malware prevention focuses on **reducing attack surfaces, detecting threats early, and limiting the impact of infections** when they do happen.

Common Malware Prevention Techniques

1. Regular System Updates and Patch Management

Many malware attacks exploit **known vulnerabilities** in operating systems and software.

Prevention measures include:

- Keeping operating systems updated
- Applying security patches regularly
- Updating applications and browsers

Example:

WannaCry exploited an unpatched Windows vulnerability, highlighting the importance of timely updates.

2. Antivirus and Endpoint Protection

Antivirus and endpoint security tools help by:

- Detecting known malware signatures
- Monitoring suspicious behavior
- Blocking malicious files before execution

While not perfect, these tools form a **first line of defense** against malware.

3. User Awareness and Safe Practices

Human error remains a major cause of malware infections.

Good practices include:

- Avoiding suspicious links and attachments
- Downloading software only from trusted sources

- Being cautious with email attachments

User awareness significantly reduces infection chances.

4. Network Security Controls

Network-based protections help limit malware spread.

These include:

- Firewalls
- Network segmentation
- Intrusion Detection and Prevention Systems (IDS/IPS)

Such controls are especially important for malware that spreads automatically.

5. Backup and Recovery Strategies

Regular backups ensure that systems can be restored in case of ransomware attacks.

Best practices:

- Maintain offline or immutable backups
- Test backup restoration procedures
- Avoid storing backups on the same network

This reduces the impact of ransomware-related data loss.

6. Principle of Least Privilege

Limiting user and application privileges reduces malware impact.

If malware runs under restricted permissions:

- System-wide damage is limited
- Critical resources remain protected

Why Prevention Is Critical

Once malware executes successfully, especially ransomware, recovery becomes difficult and costly. A layered security approach combining **technology, policies, and user awareness** provides the most effective protection.

Chapter 7: Final Observations and Conclusion

This task helped build a clear understanding of how malware behaves and why malware analysis is important in cybersecurity. By analyzing the WannaCry ransomware sample using VirusTotal, it became easier to connect theoretical malware concepts with real-world behavior.

The detection and behavior reports showed how malware is classified based on vendor detections, threat categories, and runtime actions. Observing indicators such as file encryption, system modification, and network activity helped in identifying ransomware characteristics without executing the malware.

Overall, this task highlighted the importance of behavior-based analysis and preventive security measures. Understanding malware types, their lifecycle, and common prevention techniques plays a key role in reducing the impact of malware attacks and strengthening system security.